

EE526/RA506 Machine Learning

End-Semester Exam

Date: 28 April 2025 Full Marks: 180 Time: 3 Hours

Name:

Signature

Roll Number:

Course Code

NOTE: Write your NAME (with SIGNATURE), ROLL NUMBER and COURSE CODE on the response sheet. FINAL ANSWERS to questions must be written in the space provided in the response sheet. Questions with BLANK responses will not be further evaluated. Please ATTACH the response sheet with your answer script.

1. A total of q instances are required to learn the model $\mathcal{M}$ by using *regression diagnostics* (RD) from a noisy dataset $\mathbf{S}_n$ having at most η percentage of outliers (domain expert's guess). The *Chebyshev Inequality* with threshold λ is used to reject outlier instances at each RD iteration. Provide an approximation of the number of iterations (K_a) required to remove all outliers from $\mathbf{S}_n$. [15]

$$K_a \geq \frac{\ln\left(1 - \frac{\eta}{100}\right)}{\ln\left(1 - \frac{1}{2\lambda^2}\right)}$$

2. A sensor is receiving samples of a gradually decaying signal of the form $x[n] = A\beta^n$ ($n \geq 0$, $0 < \beta < 1$). The sensor is feeding the data to a system that incrementally updates the mean as $\mu[n] = (1 - \alpha)\mu[n - 1] + \alpha x[n]$. Here, $\alpha \in (0, 1)$ is a fixed update rate and $\alpha \neq \beta$. Evaluate $\mu[n]$ for this input $x[n]$ where $\mu[0] = A$. [15]

$$\mu[n] = A \left[(1 - \alpha)^n + \left(\frac{\alpha\beta}{\alpha + \beta - 1} \right) \{ \beta^n - (1 - \alpha)^n \} \right]$$

3. The performance analysis of a binary classification problem reveals 83 *True Positives*, 22 *False Positives*, 34 *False Negatives*, and 71 *True Negatives*. Evaluate the following performance measures in *Percentage* (e.g. 49.21%) – Precision (P), Recall (R), F1 Scores (overall and individual classes), Specificity (S), False Alarm Rate (FAR), Negative Predictive Value (NPV), Accuracy (A) and Balanced Accuracy (BA). [15]

$$\mathbf{P} = 79.0476 \quad \mathbf{R} = 70.9402 \quad \mathbf{F1}^+ = 74.7748 \quad \mathbf{F1}^- = 76.9422 \quad \mathbf{F1} = 75.8585$$

$$\mathbf{S} = 89.2473 \quad \mathbf{FAR} = 10.7527 \quad \mathbf{NPV} = 67.6190 \quad \mathbf{A} = 73.3333 \quad \mathbf{BA} = 80.0937$$

4. Consider the function $\phi(x; \beta, \lambda) = \beta \frac{\lambda^x}{x!}$ where x takes discrete values from $\{0, 1, 2, \dots\}$ and $\beta, \lambda \in \mathbb{R}$. (a) Evaluate β such that $P(x|\lambda) = \phi(x; \beta, \lambda)$ is a probability distribution. (b) Deduce the expressions for $E(x)$ and $Var(x) = E[\{x - E(x)\}^2]$ where x is distributed as $P(x|\lambda)$. (c) Consider the data set $D = \{x_k : k = 1, \dots, n\}$ (x_k are i.i.d.) drawn from the mixture model $P(x|\lambda_1, \dots, \lambda_m) = \sum_{j=1}^m P(x|\omega_j, \lambda_j)P(\omega_j)$. Here, mixing parameters $P(\omega_j)$ ($j = 1, \dots, m$) are known apriori. Deduce the expression for the Maximum Likelihood Estimate of $\hat{\lambda}_j$. [2 + 3 + 4 + 6 = 15]

$$\beta = e^{-\lambda} \quad \mathbf{E}(\mathbf{x}) = \lambda \quad \mathbf{Var}(\mathbf{x}) = \lambda \quad \hat{\lambda}_j^{(t)} = \frac{\sum_{k=1}^n x_k P(\omega_j|x_k, \lambda_1^{(t-1)}, \dots, \lambda_m^{(t-1)})}{\sum_{k=1}^n P(\omega_j|x_k, \lambda_1^{(t-1)}, \dots, \lambda_m^{(t-1)})}$$

5. $S_0 = \{[-2, -1]^T, [-3, -1]^T, [-1, 1]^T\}$ and $S_1 = \{[3, 3]^T, [2, 3]^T, [4, 5]^T\}$ are i.i.d. instances from classes Ω_0 and Ω_1 , respectively. Construct the *Discriminant Function* $g(\mathbf{x})$ assuming that instances from both classes are normally distributed. [20]

$$\mathbf{g}(\mathbf{x}) = \mathbf{p}^T \mathbf{x} + q \quad \mathbf{p} = [12 \quad -4.50]^T \quad q = 1.50$$

6. Consider a SLFN with input layer $\mathbf{X} \in \mathbb{R}^2$, hidden layer $\mathbf{H} \in \mathbb{R}^3$ (*Swish* activation), and output layer $\mathbf{Y} \in (0, 1)^2$ (*SoftMax* activation). Evaluate $\mathbf{H}$ and $\mathbf{Y}$ for input $\mathbf{X} = [-5 \quad 5]^T$ and the following SLFN parameters (W_1, b_1) between $\mathbf{X} \rightarrow \mathbf{H}$, and (w_2, b_2) between $\mathbf{H} \rightarrow \mathbf{Y}$. [15]

$$W_1 = \begin{bmatrix} -0.15 & 0.20 \\ 0.11 & 0.17 \\ 0.57 & -0.77 \end{bmatrix}, \quad b_1 = \begin{bmatrix} 0.51 \\ 0.03 \\ -0.11 \end{bmatrix}, \quad W_2 = \begin{bmatrix} -0.11 & 0.13 & 0.07 \\ 0.82 & -0.13 & 0.43 \end{bmatrix}, \quad b_2 = \begin{bmatrix} 0.12 \\ 0.31 \end{bmatrix},$$

$$\mathbf{H} = [2.0465 \quad 0.1920 \quad -0.0075]^T$$

$$\mathbf{Y} = [0.1150 \quad 0.8850]^T$$

7. Consider an MLP with input $\mathbf{X} \in \mathbb{R}^D$, hidden layers $\mathbf{H}_1 \in \mathbb{R}^K$ (*Swish* activation), $\mathbf{H}_2 \in (-1, 1)^L$ (*tanh* activation), and output $\mathbf{Y} \in (0, 1)^M$ (*SoftMax* activation). The parameters between $\mathbf{X} \rightarrow \mathbf{H}_1$, $\mathbf{H}_1 \rightarrow \mathbf{H}_2$, and $\mathbf{H}_2 \rightarrow \mathbf{Y}$ are (W_1, B_1) , (W_2, B_2) , and (W_3, B_3) respectively. (a) Report the total number of trainable MLP *Parameters* and *Feedforward FLOPs* (excluding activation evaluation). (b) Deduce the following for weight update by gradient descent while minimizing the *Cross-Entropy Loss* $\mathcal{L}_{CE}$ over the training dataset $\mathbf{D}_{trn} = \{(\mathbf{X}_n, \mathbf{Y}_n); n = 1, \dots, N\}$. [5 + 15 = 20]

$$\mathbf{Params} = (KD + LK + ML) + (K + L + M) \quad \mathbf{FLOPs} = 2(KD + LK + ML) + (K + L + M)$$

$$\frac{\partial \mathcal{L}_{CE}}{\partial W_1(i, j)} =$$

8. Perform kernel-based regression for the multivariate function $\mathbf{y} = \mathbf{F}(\mathbf{x})$, where $\mathbf{x}, \mathbf{y} \in \mathbb{R}^2$. The following table lists 10 data instances $(\mathbf{x}, \mathbf{y})$. Using these instances, report the estimate of $\mathbf{y}$ at $\mathbf{x} = [0.52 \quad 0.40]^T$. Use an Epanechnikov kernel with bandwidth $h = 0.19$. [15]

	Sample index									
	1	2	3	4	5	6	7	8	9	10
x_1	0.62	0.40	0.57	0.49	0.39	0.58	0.75	0.44	0.46	0.41
x_2	0.35	0.19	0.32	0.61	0.54	0.29	0.43	0.30	0.63	0.37
y_1	0.22	0.02	0.01	-0.04	-0.04	0.15	1.42	0.28	-0.50	-0.01
y_2	0.43	0.08	0.01	-0.16	-0.55	-0.36	-1.12	-0.75	-0.35	0.26

$$\mathbf{y} = [0.1211 \quad -0.0499]^T$$

9. The instances in the multi-category dataset $\mathbf{S}_m$ consisting of 5 different classes are present in the proportion of 10 : 14 : 7 : 5 : 20. Report the *Misclassification* (H_m), *Entropy* (H_e) and *Gini* (H_g) impurities of dataset $\mathbf{S}_m$. [10]

$$\mathbf{H}_m = 0.6429$$

$$\mathbf{H}_e = 2.1605$$

$$\mathbf{H}_g = 0.7545$$

10. The dataset $S = S^+ \cup S^-$ is presented to a node of a classification tree. Here, *Positive Class* $S^+ = \{(10, 8), (-7, -2), (-2, 6), (-3, 9), (2, 2)\}$ and *Negative Class* $S^- = \{(10, -5), (12, 3), (1, -6), (8, 2), (9, -5)\}$. Split this dataset S by maximizing the misclassification impurity drop. Report (a) the optimal decision rule in the form $X[j^*] \geq \theta^*$ (threshold θ is taken from data values); and (b) resultant impurity drop ΔH . [20]

$$j^* =$$

$$\theta^* =$$

$$\Delta H =$$

11. Group the dataset $D = \{3, 31, 35, 6, 63, 7, 65, 38, 4, 16\}$ into $K = 3$ clusters by K-means clustering. Iterate till convergence and report the data labels (L_t) and cluster centroids (μ_1, μ_2, μ_3) for each iteration t in the following table. [20]

Data $\rightarrow$	3	31	35	6	63	7	65	38	4	16	μ_1	μ_2	μ_3
L_0	1	2	3	1	2	3	1	2	3	1			
L_1													
L_2													
L_3													
L_4													
L_5													
L_6													